

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

IV Semester of M. Sc. (Biochemistry) Examination, April, 2019

Course code & Name BC 863 & Clinical Biochemistry

Date: 29/04/2019 Day: Monday Time: 1.30 PM to 1.50 PM Maximum Marks: 10

MCQ

Important Instructions:

- Tick the correct answer and it should be written in question paper itself.
- Use of non-programmable calculator is allowed.

Q - I	Choose the correct answer for the following questions.		10
1.	The first condition identified as an “inborn error of metabolism” was		01
	(i) Alcaptonuria	(ii) Albinism	
	(iii) Phenylketonuria	(iv) Tyrosinemia	
2.	The key regulatory enzyme of cholesterol synthesis is-		01
	(i) HMG-CoA synthase	(ii) Mevalonate kinase	
	(iii) HMG-CoA reductase	(iv) HMG-CoA lyase	
3.	The L-aminoacids are absorbed from intestine by		01
	(i) Active transport	(ii) Pinocytosis	
	(iii) Passive diffusion	(iv) Facilitated diffusion	
4.	A 23-year-old woman has been fasting for religious reason for several days. Her brain has reduced its need for glucose by using which of the following substances as an alternate source of energy?		01
	(i) Beta hydroxyl butyrate	(ii) Glycogen	
	(iii) Fatty acids	(iv) Glycerol	
5.	An enzyme used in both Glycolysis and Gluconeogenesis is:		01
	(i) Glucose 6-phosphatase	(ii) 3-phosphoglycerate kinase	
	(iii) Hexokinase	(iv) Phosphofructokinase-1	

6.	An intermediate of the citric acid cycle that undergoes reductive amination with glutamine as nitrogen donor is		01
	(i) α -ketoglutarate	(ii) Glutamine	
	(iii) NADPH	(iv) H^+	
7.	Which of the following is a common compound shared by the TCA cycle and the Urea cycle?		01
	(i) α -ketoglutarate	(ii) Oxaloacetate	
	(iii) Succinyl coA	(iv) Fumurate	
8.	Which of the following lipid lowering drugs act by inhibiting the absorption of dietary cholesterol?		01
	(i) Statins	(ii) Colestipol	
	(iii) Ezetimble	(iv) Clolestyramine	
9.	Which of the following vitamins can be used in high doses to treat hypercholesterolemia?		01
	(i) Thiamine	(ii) Folic acid	
	(iii) Niacin	(iv) Riboflavin	
10.	The hormones aldosterone and ADH both have an important function in:		01
	(i) fluid balance in the body	(ii) the regulation of acid concentration in the body	
	(iii) stimulation of a conscious desire for water	(iv) the activity of buffer systems	

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IV Semester of M. Sc. (Biochemistry) Examination, April, 2019

Course code & Name BC 863 & Clinical Biochemistry

Date: 29/04/2019 Day: Monday Time: 1.51 PM to 3.30 PM Maximum Marks: 25

Instructions:

1. Section I and II must be attempted in SINGLE ANSWER SHEET.
2. Make suitable assumptions and draw neat figures wherever required.
3. Use of a non - programmable calculator is allowed.
4. Show necessary calculations.

SECTION -I		
Q – II Answer the following questions as directed (Any 5)		10
1.	What is viral hepatitis? Discuss briefly.	02
2.	How sodium concentration is regulated or balanced in the body? Justify your answer.	02
3.	Why is it important for the amount of water intake to equal the amount of water output?	02
4.	Write a short note on Hypovolemia and Hypervolemia.	02
5.	Write a short note on different types of Dialysis.	02
6.	Write a short note on polygenic hypercholesterolemia.	02
7.	Write a short note on Budd Chiari Syndrome.	02
Section-II		
Q-III Answer the following questions as directed (attempt any 5)		15
1.	What are complications of acute and chronic diabetes mellitus? Discuss briefly.	03
2.	What is Nephrotic syndrome? What are the symptoms and treatments for Nephrotic syndrome? Discuss briefly.	03
3.	Write a short note on Girke's disease.	03
4.	What is Galactosemia? What are the different types of Galactosemia? Discuss briefly.	03
5.	What is hyperlipidemia? Describe briefly the causes for secondary hyperlipidemia?	03
6.	Write a short note on uremia.	03

7.	What are the different types of autoimmune liver disease? Describe briefly any of them.	03
8.	Write a short note on Automatization & Expert system in Laboratory Medicine.	03